Audrey Oliver

+1(650)773-2946 akoliver@sdsu.edu linkedin.com/in/akoliver My Webpage

Profile

- Computational Science PhD student interested in mathematical modeling of infectious diseases, with a focus on vector-borne transmission and healthcare accessibility.
- Development of simulation frameworks that bridge biology, mathematics, and data science to support public health decision-making.
- Use of machine learning, statistical inference, and numerical methods to analyze epidemiological data and guide intervention strategies.
- Research at the intersection of computational biology, applied dynamical systems, and global health equity.

Education

Doctor of Philosophy in Computational Science **08/2024 - 05/2029 (*Anticipated*)**
University of California, Irvine & San Diego State University *GPA: 3.86*

Master of Science in Computational Science **08/2022 - 05/2024**
San Diego State University *GPA: 3.63*

Bachelor of Science in Biology, Minor in Mathematics **08/2017 - 12/2021**
San Diego State University *GPA: 3.5*

Research Experience

Graduate Research Assistant **08/2023 – Present**
SDSU Disease Modeling (DiMo) Lab *San Diego, CA*

- Mentor undergraduate researchers in computational disease modeling, programming, and interpretation of research findings.

Research Projects

Dengue Surveillance Reconstruction and Transmission Estimation **01/2026 – Present**
SDSU Disease Modeling (DiMo) Lab

- Develop methods to reconstruct daily dengue incidence from sparse surveillance reports using random forest models and daily climate data.
- Evaluate reconstruction methods against observed daily incidence in Taiwan to assess how reporting intervals affect accuracy and temporal patterns.
- Estimate time-varying reproduction numbers using a human–vector transmission framework and quantify uncertainty through profile likelihood confidence intervals.

Spatial Healthcare Accessibility for Dengue Fever in Nepal **09/2024 - Present**
SDSU Disease Modeling (DiMo) Lab

- Integrate monthly dengue case counts, healthcare facility locations, population data, and environmental datasets to assess healthcare accessibility across 77 districts in Nepal.
- Implement floating catchment area methods in R using travel times between district population-weighted centroids and healthcare facilities.
- Compare walking and mixed transportation scenarios and evaluate the effects of precipitation-adjusted travel speeds on estimated accessibility.

Modeling the Pharmacodynamics of ART on Resistance in Latent Infection of HIV **06/2024 - Present**
SDSU Disease Modeling (DiMo) Lab

- Develop a mathematical model incorporating drug-resistant HIV, latently infected cells, and antiretroviral pharmacodynamics to investigate treatment effects on viral control.
- *Manuscript for publication in progress*

Modeling the Spatiotemporal Distribution of HIV Infection in the Brain **09/2022 - 09/2023**
SDSU Disease Modeling (DiMo) Lab

- Create compartmental model for the distribution of HIV among different area of the brain, and perform varying analyses to determine influence of parameters. Compute reproduction number of HIV within each compartment considered.

Modeling the Pharmacodynamics of ART for HIV in the Brain **04/2021 - 10/2023**
SDSU Disease Modeling (DiMo) Lab

- Developed and simulated a mathematical model of HIV infection in the brain to compare predicted viral suppression under single-drug, two-drug, and three-drug antiretroviral regimens.
- Built an interactive Tableau dashboard to visualize predicted treatment responses across user-selected drug combinations.

Instructional and Mentoring Experience

Teaching Associate for MATH340 **01/2024 – Present**

San Diego State University, Dept. of Math & Stats *San Diego, CA*

- Lead lab sections for upper division undergraduate course, *Programming in Math.*
- Introduce Python skills such as data visualization using Matplotlib, numerical computing and vectorization using NumPy arrays and vectors, symbolic math and computer algebra with SymPy.
- Create weekly, supplementary activities covering a variety of topics: sequences, series, and their representation using numerical arrays, and numerical methods such as error analysis, growth rate analysis, differentiation, integration, interpolation, root finding.
- Grade assignments, provide constructive feedback to improve coding skills and understanding of numerical methods.

Scholarships and Awards

ASSICS Scholarship **08/2022 – 05/2025**

SDSU Computational Science Research Center

- $10,000 per year for graduate students interested in pursuing academic work and research in Interdisciplinary Computational Science and Engineering topics.
- Academic support program for professional development and exploration of career pathways to becoming interdisciplinary scientists and engineers.
- Supported by National Science Foundation grant and housed in the SDSU's Computational Science Research Center.

Raymond Moberly Outstanding Service Award **03/2025**

CSRC ACSESS Event 2025

- Recognized for exceptional service and mentorship in the Computational Science and Engineering community at SDSU, honoring academic excellence and impactful student and community engagement.

CSRC Alumni Award **03/2025**

CSRC ACSESS Event 2025

- Present Master's Research to SDSU faculty and CSRC collaborators from National Labs and Industry.
- Judged by multiple faculty/ collaborators from industry for five minute video presentation, one minute overview of research in front of audience, and poster presentation.
- Awarded for outstanding performance and dedication to research.

Award for Women Engineers/Scientists **03/2024**

CSRC ACSESS Event

- Present Master's Research to SDSU faculty and CSRC collaborators from National Labs and Industry.
- Judged by multiple faculty/ collaborators from industry for five minute video presentation, one minute overview of research in front of audience, and poster presentation.
- Awarded for outstanding performance and dedication to research as a woman in engineering/science.

Best Poster Presentation Award **04/2023**

Southern California Applied Mathematics Symposium

- Present research to faculty, students, and general symposium attendees with interests in applied mathematics.
- 1 of 3 students to receive an award for best poster, as judged by other symposium attendees.

Leadership & Involvement

Society for Industrial & Applied Mathematics, SDSU Chapter **08/2022 – Present**

Club Board Member

- **Treasurer (June 2024-Present):** Handle all financial affairs and budgeting of the organization, maintain all necessary accounting records, and prepare monthly financial reports for the membership. Collect and deposit all club dues and fees. Maintain bank accounts in the organization's name, requiring signatures of both the Treasurer and President for authorized disbursements. Secure funding for students to attend conferences.
- **Public Relations Chair (June 2023-May 2024):** Responsible for all promotional and community-related organization correspondence, such as maintaining the club website, develop and distribute press releases, newsletters, brochures, and shall keep copies thereof on file.

Presentations

- Framework for Data-Driven Modeling to Estimate Reproduction Numbers, the Most Critical Epidemic Thresholds, of Air-Borne and Vector-Borne Diseases. CSRC ACSESS Symposium, San Diego, CA, May 2026.
- Framework for Data-Driven Modeling to Estimate Reproduction Numbers, the Most Critical Epidemic Thresholds, of Air-Borne and Vector-Borne Diseases. Joint Mathematics Meeting, Washington, D.C., Jan 2026.
- Framework for Data-Driven Modeling to Estimate Reproduction Numbers, the Most Critical Epidemic Thresholds, of Air-Borne and Vector-Borne Diseases. CSRC ACSESS Symposium, San Diego, CA, March 2025.
- Modeling the Spatiotemporal Distribution of HIV Infection in the Brain. Southern California Applied Mathematics Symposium, San Diego, CA, April 2024.
- Mathematical Models for HIV Infection in the Brain: Implications for Treatment Regimens. CSRC ACSESS Symposium, San Diego, CA, March 2024.
- Mathematical Models for HIV Infection in the Brain: Implications for Treatment Regimens. SDSU Student Research Symposium, San Diego, CA, March 2024.
- Mathematical Models for HIV Infection in the Brain: Implications for Treatment Regimens. Joint Mathematics Meeting, San Francisco, CA, January 2024.
- Modeling the Spatiotemporal Distribution of HIV Infection in the Brain. Predictive Modeling in Biology and Medicine Conference, Riverside, CA, November 2023.
- Modeling the Spatiotemporal Distribution of HIV Infection in the Brain. Southern California Applied Mathematics Symposium, Irvine, CA, April 2023.
- Modeling the Spatiotemporal Distribution of HIV Infection in the Brain. CSRC ACSESS Symposium, San Diego, CA, April 2023.
- Modeling the Spatiotemporal Distribution of HIV Infection in the Brain. SDSU Student Research Symposium, San Diego, CA, March 2023.
- Modeling Cluster Analysis of Antiretroviral Therapy for HIV Infection in the Brain. SDSU Student Research Symposium, San Diego, CA, March 2022.

Technical Skills

Programming: R, Python, MATLAB, C

Epidemiologic Analysis: Surveillance data analysis, daily incidence reconstruction, time-varying reproduction number estimation, uncertainty quantification

Statistics & Machine Learning: Random forests, parameter estimation, model validation, principal component analysis, cluster analysis

Mathematical Modeling & Numerical Methods: Compartmental models, ordinary and partial differential equations, antiretroviral pharmacodynamics, numerical solution of differential equations, sensitivity analysis

Spatial Analysis: Raster and vector data processing, geospatial data integration, healthcare accessibility analysis, travel-time modeling

Data Visualization: Matplotlib, Tableau, spatial mapping

Research Tools: Git/GitHub, Jupyter Notebook, LaTeX

Professional Affiliations

Society For Industrial and Applied Mathematics
Society for Mathematical Biology
American Mathematical Society